

6.10 Similar Figures

Lesson Introduction

 Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.		 Learning Target	I can solve mathematical problems using corresponding angles and sides of similar figures.	
 Benchmark Coherence	Building On MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.			Working Toward N/A	
 Essential Questions	- How can similar figures be used to find the measures of angles? - How can similar figures be used to find the measures of sides?				
 Lesson Narrative	This lesson will apply the relationships of similar triangles and corresponding components from Lessons 1-9 to other similar polygons. Students will determine the measures of angles through congruence of corresponding parts and the measures of side lengths through proportional reasoning.				
 Component Summary	6.10.1 Warm-Up (~5 min.)	Proving two triangles similar (SE p. 349)	6.10.3 Your Turn! (15-20 min.)	Determining the measures of corresponding parts of similar figures (SE p. 351)	
	6.10.2 Collaboration (15-20 min.)	Connecting relationships of similar triangles to similar polygons (SE p. 349)	6.10.4 Wrap-Up (~5 min.)	Reflection on the lesson understanding (SE p. 353)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similarity - Congruent - Proportional - Corresponding		N/A		(Optional) Consider cutting out the images and preimages from 6.10.2 Collaboration and 6.10.3 Your Turn! for students who may need the differentiated support of being able to manipulate the figures when determining corresponding sides and angles.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may match the corresponding parts in the similarity statement incorrectly. - Students may name the scale factor of ΔHRA to ΔVKM instead of from ΔVKM to ΔHRA in 6.10.2 Collaboration.	Consider offering 6.10.3 Your Turn! as a partner or individual component.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 6.10.2 Collaboration is organized to be completed with a partner. Begin this activity individually, connect with the partner, and determine the best answer. Finally share with the class what has been agreed upon. You can continue to use the Think-Pair-Share routine into 6.10.3 Your Turn!, if desired.	MA.K12.MTR.1.1: Participate in Effortful Learning 6.10.2 Collaboration and 6.10.3 Your Turn! both require students to participate fully in effortful learning both with partners and independently. Students must analyze the problems and ask questions when they are in small groups. Students also will build perseverance and remain engaged while working independently.
 Differentiate Instruction	Students may struggle with identifying the corresponding parts of similar figures. Consider providing cut out images of the figures that students can manipulate to determine the corresponding parts. After the figures have been oriented with each other, have the students complete or make a list of the corresponding sides and angles.	
Lesson Notes:		

6.10.1 Warm-Up: Bell Work

Component Overview: Students prove that two triangles are similar by completing the missing statements and reasons in a partially completed two-column proof.

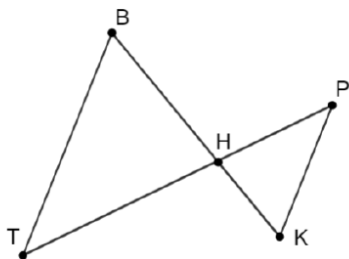


6.10.1 Warm-Up: Bell Work

1. Complete the proof.

Given: $\overline{BT} \parallel \overline{PK}$

Prove: $\triangle BHT \sim \triangle KHP$



Statement	Reason
1. $\overline{BT} \parallel \overline{PK}$	1. Given
2. $\angle TBH \cong \angle PKH$	2. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
3. $\angle BTH \cong \angle KPH$	3. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
4. $\triangle BHT \sim \triangle KHP$	4. Angle-Angle Similarity



Students may choose to use one of the alternate interior angle relationships and the vertical angles relationship. Both justifications are correct. Students should be encouraged to use any valid reasoning.

6.10.2 Collaboration: Similar Figures

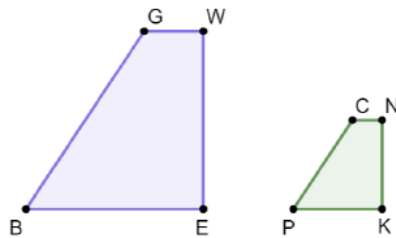
Component Overview: Students evaluate similar figures and determine unknown corresponding angles and corresponding sides.



6.10.2 Collaboration: Similar Figures

With your partner, answer the following questions.

1. $GWEB$ and $CNKP$ are shown.



- If $GWEB \sim CNKP$, what is true about the corresponding angles?
Corresponding angles are congruent.
- If $GWEB \sim CNKP$, what is true about the corresponding sides?
Corresponding sides are proportional.
- Fill in the table given that $GWEB \sim CNKP$.

Corresponding Angles	Corresponding Sides
$\angle GBE \cong \angle CPK$	$\frac{PK}{BE} = \frac{KN}{EW}$
$\angle PKN \cong \angle BEW$	$\frac{CN}{GW} = \frac{KN}{EW}$
$\angle PCN \cong \angle BGW$	$\frac{CP}{GB} = \frac{NC}{WG}$
$\angle GWE \cong \angle CNK$	$\frac{KP}{EB} = \frac{PC}{BG}$



Think-Pair-Share

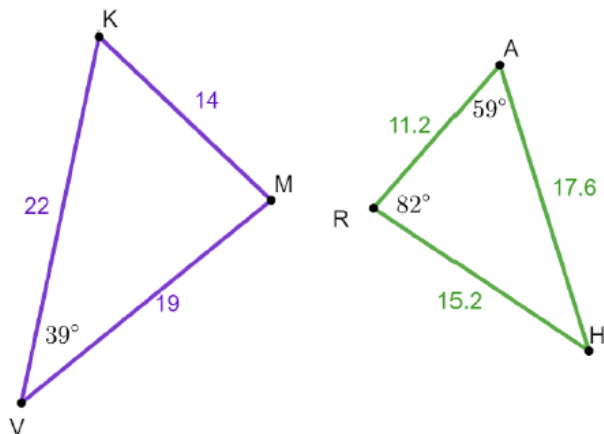
This section is intentionally designed for partner work. Consider reviewing group norms that may begin with independent thought, then lead into partnered discussion. Finally, follow up with class shared responses.



- What is different between questions 1a and 1b?
- Does the word “congruent” have the same meaning as “proportional”?
- What is the difference between congruent and proportional?

6.10.2 Collaboration: Similar Figures (continued)

2. $\triangle VKM$ and $\triangle HRA$ are shown. $KM = 14$, $MV = 19$, $VK = 22$, $RA = 11.2$, $AH = 17.6$, $HR = 15.2$, $m\angle KVM = 39^\circ$, $m\angle HRA = 82^\circ$, and $m\angle RAH = 59^\circ$.



- Using the given information, what can you conclude about $\triangle VKM$ and $\triangle HRA$?
The side lengths of triangles VKM and HAR are proportional.
- What similarity theorem or postulate could be used to prove the triangles are similar?
SSS Similarity
- Write a similarity statement for $\triangle VKM$ and $\triangle HRA$.
 $\triangle VKM \sim \triangle HAR$



For question 2a, the order of vertices was purposely not aligned so that students cannot rely on that to determine corresponding parts.

For question 2b, students may also determine the $\angle RHA$ and then use SAS similarity. Consider having students who used different methods share with the whole group.

6.10.2 Collaboration: Similar Figures (continued)

d. Complete the table.

Angle Measures	Side Proportions
$m\angle VKM = 59^\circ$	$\frac{MK}{RA} = \frac{14}{11.2}$
$m\angle KMV = 82^\circ$	$\frac{MV}{RH} = \frac{19}{15.2}$
$m\angle RHA = 39^\circ$	$\frac{KV}{AH} = \frac{22}{17.6}$

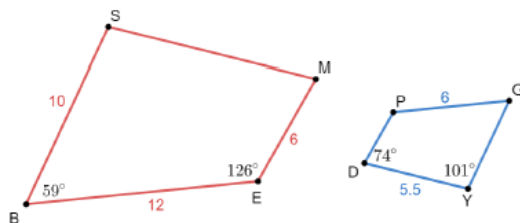
e. What scale factor are the sides of $\triangle VKM$ multiplied by to find the corresponding sides of $\triangle HRA$?

0.8 or $\frac{4}{5}$

f. If $MK = 21$, what should the length of $\overline{RA}$ be so that the triangles are similar?

16.8

3. $SMEB$ and $YDPG$ are shown. $SB = 10$, $BE = 12$, $EM = 6$, $DY = 5.5$, $PG = 6$, $m\angle SBE = 59^\circ$, $m\angle BEM = 126^\circ$, $m\angle PDY = 74^\circ$, and $m\angle DYG = 101^\circ$.



a. If $SMEB \sim YDPG$, determine the measure of each.

$$\begin{aligned} m\angle BSM &= 101^\circ & m\angle SME &= 74^\circ \\ m\angle DPG &= 126^\circ & m\angle PGY &= 59^\circ \\ SM &= 11 & PD &= 3 & GY &= 5 \end{aligned}$$

b. Explain how the similarity relationship, $SMEB \sim YDPG$, helped determine which sides of the quadrilaterals are proportional.

Answers will vary. The similarity statement tells me what angles are congruent. From this I can find corresponding sides that are proportional.



If students respond to question 2e with 1.25, then they incorrectly named the scale factor of $\triangle HRA$ to $\triangle VKM$ instead of $\triangle VKM$ to $\triangle HRA$.

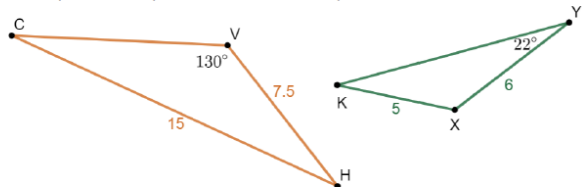
6.10.3 Your Turn!

Component Overview: Students find unknown side lengths and angle measures using similar shapes and corresponding parts.



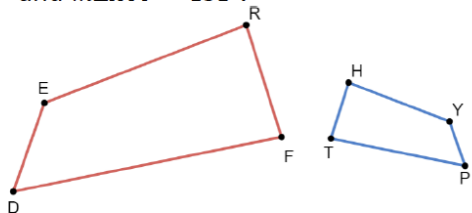
6.10.3 Your Turn!

1. $\triangle CVH$ and $\triangle YXK$ are shown where $VH = 7.5$, $CH = 15$, $KX = 5$, $XY = 6$, $m\angle CVH = 130^\circ$, and $m\angle XYK = 22^\circ$.



- Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each angle.
 $m\angle VCH = 22^\circ$ $m\angle VHC = 28^\circ$
 $m\angle YKX = 28^\circ$ $m\angle YXK = 130^\circ$
- Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each side.
 $CV = 9$ $KY = 10$

2. $RFDE$ and $HTPY$ are shown, where $RE = 6.4$, $ED = 2.8$, $HY = 3.2$, $HT = 1.7$, $PT = 4$, $m\angle ERF = 86^\circ$, $m\angle EDF = 59^\circ$, and $m\angle HYP = 131^\circ$.



Assume: $RFDE \sim HTPY$

- Find the measure of $\angle DER$. Show your work or explain your thinking.
 $m\angle DER = 131^\circ$ because the similarity statement tells me that $\angle DER$ corresponds to $\angle PYH$ and the $m\angle PYH = 131^\circ$.



If students struggle with seeing the corresponding parts from the figure, consider scaffolding question 1 by first asking students to complete a list of all corresponding angles and sides by using the order of the vertices in the similarity statement. For example, use the letters in the similarity statement to determine the following correspondence.

- $\angle CVH$ corresponds to $\angle$ _____.
- $\angle CHV$ corresponds to $\angle$ _____.
- $\angle VCH$ corresponds to $\angle$ _____.
- $\overline{CV}$ corresponds to _____.
- $\overline{CH}$ corresponds to _____.
- $\overline{HV}$ corresponds to _____.

6.10.3 Your Turn! (continued)

- b. Determine each measure.

$$m\angle HTP = \underline{84^\circ} \quad m\angle DFR = \underline{84^\circ}$$

$$m\angle THY = \underline{86^\circ} \quad m\angle YPT = \underline{59^\circ}$$

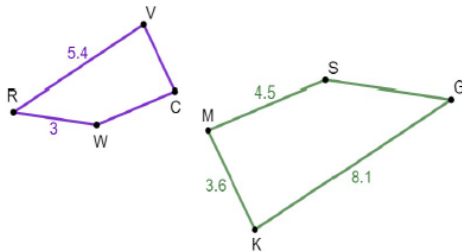
- c. What is the ratio of the corresponding sides of $RFDE$ to $HTPY$?

The ratio is 2:1.

- d. Determine each measure.

$$DF = \underline{8} \quad RF = \underline{3.4} \quad YP = \underline{1.4}$$

3. Kirstyn and Roberto were finding the missing side measures for $RVCW$ and $GKMS$. They were given $RVCW \sim GKMS$, $VR = 5.4$, $RW = 3$, $SM = 4.5$, $MK = 3.6$, and $KG = 8.1$.



Kirstyn's and Roberto's work to find the measure of $\overline{VC}$ is shown.

Kirstyn's Work	Roberto's Work
$\frac{8.1}{5.4} = 1.5$	$\frac{8.1}{5.4} = 1.5$
$3.6 \times 1.5 = 5.4$	$3.6 \div 1.5 = 2.4$
$VC = 5.4$	$VC = 2.4$

- a. Who do you agree with?

I agree with Roberto.

- b. Explain the error in the incorrect work.

Kirstyn multiplied the 1.5 when she should have divided.

- c. Complete the statements.

$$CW = \underline{3} \quad SG = \underline{4.5}$$



If students don't remember the sum of the angles of a quadrilateral, spiral back to decomposing figures into triangles.



For students who need an extra challenge, task them with creating a problem with another polygon other than a quadrilateral.

6.10.4 Wrap-Up: Reflection

Component Overview: *Students reflect on their progress throughout the lesson.*



6.10.4 Wrap-Up: Reflection

1. One thing I struggled with today was ...
Answers will vary.
2. One thing I need to practice more is ...
Answers will vary.
3. One thing I understand very well is ...
Answers will vary.



Consider keeping a list of “One thing I struggled with” statements that continue reoccurring. If patterns emerge, consider reteaching these topics.

Individually praise students for the topics they understand. Consider inviting them to be an area expert on that topic for a future lesson.